

Unit 2: Business Data Visualization

Assignment 5: Data Visualization for Business Data Analytics using Python

Topics Covered: Matplotlib and Seaborn for Data Visualization – Line Chart, Bar chart, Pie chart, Histogram, Box plot, Scatter plot, Funnel Chart (plotly), Subplot

The assignment aims to provide a comprehensive guide on the data journey flow, focusing on various visualization types like bar charts, pie charts, histograms, box plots, scatter plots, funnel charts, subplots.

Data Preparation

Dataset Selection: Choose a business-related dataset (e.g., sales data, customer information, marketing metrics, etc.) available in CSV, Excel, or any other suitable format.

Data Import: Use Pandas to read and prepare the dataset for visualization.

Data Visualization

Matplotlib and Seaborn: Create various types of plots (line plot, bar plot, histogram, scatter plot) using Matplotlib and Seaborn on business data (e.g., sales trends, customer distribution). Seaborn is great for quickly creating visually appealing plots with minimal code, while Matplotlib offers more customization options and fine-grained control. Seaborn is often considered better than Matplotlib for statistical data visualization.

Visualization Customization: Customize visualizations by adding labels, titles, legends, and styling.

Plotly is open-source, browser-based graphing library for Python. It provides unique interactivity feature, which is not seen in other visualization libraries.

A subplot allows multiple graphs to be displayed within a single figure, enabling easy comparison of related data. It is commonly used to create dashboards that present different insights together in a compact visual layout.

Line Chart: A line chart is suitable for visualizing the trend in sales over months or years, helping businesses identify patterns, overall growth.

- Display a line chart representing trend in sales over years.
- Use Case for Practice: Tracking Sales Over Time

Bar Chart: Bar charts are effective for comparing sales figures of different products, providing a clear visual representation of performance and helping in decision-making.

- Display a bar chart representing sales figures per month or any relevant business metric.
- Use Case: Comparing Sales Performance Across Products

Pie Chart: A pie chart can represent the market share of different product categories, making it easy to understand the distribution and identify dominant segments.

- Illustrate the distribution of product categories or any categorical variable in the dataset.
- Use Case: Display Market Share

Histogram: Histograms are suitable for visualizing the age distribution of customers, enabling businesses to target specific age groups in marketing strategies.

- Create a histogram showing the distribution of customer ages or any numerical variable.
- Use Case: Analyze Customer Age Distribution

Box Plot: Boxplots help in identifying salary outliers within an organization, aiding HR departments in understanding salary distributions and potential anomalies.

- Generate a box plot depicting the distribution of revenue across different regions or product types.
- Use Case: Identifying Outliers in Employee Salary

Scatter Plot: Scatter plots are useful for exploring correlations, such as understanding how changes in advertising expenditure relate to variations in sales.

- Plot a scatter plot showcasing the relationship between advertising expenditure and sales figures.
- Use Case: Examine the Relationship Between Advertising Expenditure and Sales

Funnel Chart (Optional - Requires Additional Libraries like Plotly): Funnel charts are suitable for visualizing the sales conversion process, allowing businesses to identify bottlenecks and optimize their sales funnel.

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- Create a funnel chart to visualize the sales conversion process through different stages.
- Use Case: Analyze Sales Conversion Rates

Interpretation and Insights

Interpretation: Describe and interpret the insights derived from each visualization.

Comparison: Discuss the effectiveness of each visualization type in conveying specific types of information.

Business Relevance: Explain how these visualizations can aid in business decision-making or strategy development.

Students are required to create a folder as FourDigitRollno_Roomno_DeptCode_BDA (For Example, 0012_14_COMA_BDA) on their desktop. All the files related to the assignment should be saved within this folder. Students are required to use either Spyder or Jupyter Notebook to complete the assignment. The final working code should be saved as AssignmentNo_Rollno.py

Sales Trend Analysis (Line Chart)

1. A mid-sized retail company wants to evaluate its sales growth pattern over the last five years. The annual sales data (in USD) is given below:

2021: 55,000
2022: 60,000
2023: 65,000
2024: 70,000
2025: 75,000

- Use matplotlib to display a line chart with 'o' marker representing the sales trend over these 5 years.
- Customize the chart: Set the color of the plot to purple, title of the plot, labels for x-axis and y-axis.
- Use seaborn to display a line chart representing the sales trend over these 5 years.

Product Performance Comparison (Bar Chart)

2. A retail company wants to compare monthly sales performance across different products.

Product	Sales (in USD)
Laptop	5000
Phone	7000
Tablet	3000
Monitor	4000
Headset	2000

- Use matplotlib or seaborn to display a bar chart.
- Customize the bar chart.

Market Share Analysis (Pie Chart)

3. A company wants to understand the market share contribution of different product categories. The following data represents the total sales (in USD) for each category over a year:

Category	Sales (in USD)
Electronics	40,000
Furniture	25,000
Clothing	15,000
Beauty	10,000
Sports	10,000

- Plot a pie chart illustrating the market share of each category using matplotlib.
- Customize the chart.

Customer Demographics Analysis (Histogram)

4. A retail company wants to analyze the age distribution of its customers to design targeted marketing campaigns. The dataset below includes the ages of customers.
22, 25, 29, 34, 36, 22, 23, 45, 39, 42, 41, 28, 27, 25, 30, 31, 22, 21, 33, 32, 29, 26, 35, 40, 38, 23, 22, 24, 26, 28, 25, 24, 30, 33, 34, 31, 29, 40, 45, 42, 21, 22, 23, 27, 28, 36, 37, 39, 25, 26, 27, 30, 31, 22, 21, 29, 28, 32, 33, 41, 42, 35, 34, 36, 37, 38, 39, 40, 41, 45, 25, 26, 28, 29, 30, 31, 32, 33, 21, 22, 23, 24, 27, 28, 29, 30, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45
- Use matplotlib or seaborn to plot a histogram of customer ages.
 - Customize the histogram with appropriate title, labels for x-axis and y-axis, color, Bins to group ages (e.g., 20–25, 26–30,).

Salary Distribution & Outlier Detection (Box Plot)

5. An organization wants to analyze salary distribution across departments to detect pay inconsistencies and potential outliers. The salaries (in thousands of dollars) for employees across three departments are as follows:

Department	Salaries (\$K)
Sales	50, 55, 60, 62, 65, 70, 120
IT	70, 75, 80, 85, 90, 95, 100
HR	40, 45, 50, 55, 60, 62, 65

- Use matplotlib or seaborn to create a box plot comparing salary distribution across departments.
- Customize the plot.

Advertising Spend vs Sales Relationship (Scatter Plot)

6. A company wants to assess whether increased advertising expenditure leads to higher sales. The following data represents the monthly advertising expenditure (in thousands of dollars) and corresponding sales figures (in thousands of units):

Advertising Expenditure (\$K)	Sales (Units \$K)
10	15
15	25
20	35
25	45
30	55
35	60
40	65
45	70
50	75

- Plot a scatter plot using Matplotlib or Seaborn.
- Customize the plot.

Sales Funnel Analysis (Funnel Chart with Plotly in Google Colab)

7. A sales manager wants to visualize customer drop-offs across different sales stages to identify bottlenecks. Create a Pandas DataFrame with sample data. It represents a funnel with stages ('Sent', 'Received', 'Clicks', 'Sales') and corresponding percentages (100, 70, 30, 10).

- Plot a Funnel chart using Plotly Express.
- Set 'percentage' on the x-axis, and 'stage' on the y-axis.

8. Business Dashboard Creation (Subplot)

A retail company CityMart tracks its monthly business performance across sales, customers, product categories, and marketing effectiveness. Management wants to build a compact visual dashboard to support quick and effective managerial decision-making.

The following data are provided for the year 2025.

- Monthly sales values ranging from \$30,000 to \$80,000
- Monthly customer count between 500 and 1,800
- Product category contribution:

- Electronics – 40%
- Clothing – 25%
- Groceries – 20%
- Accessories – 15%
- Monthly marketing spend ranging from \$2,000 to \$12,000 and corresponding revenue between \$30,000 and \$80,000
- a. **Data Preparation**
 - i. Create Python lists for months, monthly sales, customer counts, marketing spend, revenue, and product category contributions.
 - ii. Combine the data into a Pandas DataFrame.
 - iii. Display the first five records of the DataFrame.
- b. **Business Dashboard Visualization (Matplotlib – Subplots)**
 - i. Create a 2×2 dashboard using Matplotlib subplots:
 - Line Chart: Monthly sales trend with markers
 - Bar Chart: Monthly customer count
 - Pie Chart: Product category contribution (%)
 - Scatter Plot: Relationship between marketing spend and revenue
 - ii. Customize the figure by adding titles, axis labels, legends, gridlines, and percentages where appropriate.
 - iii. Add an overall dashboard title:
“CityMart Business Dashboard – 2025”
 - iv. Adjust layout using `plt.tight_layout()` and display the dashboard.

Exercise:

Analyze data recorded over time to understand its structure and behavior by plotting time series graphs to identify long-term trends, detect seasonal patterns, and highlight anomalies or unusual fluctuations.

Stock Price Trend Evaluation (Time Series Analysis)

A financial analytics firm is assigned to study the historical stock price movement of a publicly listed company. The objective is to understand daily price fluctuations, identify long-term trends, and generate insights that can support investment decisions and portfolio strategy. The firm has been provided with a dataset named `stock_data.csv`, which contains daily stock prices along with corresponding dates.

- Import the required Python libraries—pandas, numPy, matplotlib, and seaborn—for data manipulation and visualization.
- Load the dataset `stock_data.csv` using the `read_csv()` function from the Pandas library.
- Convert the Date column into DatetimeIndex format by using the `parse_dates` parameter to enable time-series operations.
- Create a line chart to visualize the daily stock price movement over time, helping analysts observe short-term volatility and overall price behavior.
- Resample the stock price data on a monthly basis and plot a line chart to visualize long-term trends, smoothing daily fluctuations to support strategic investment analysis.